Group 6
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MEMORY METER

PEOPLE MOVE MEMORIES STAY

Workbook

Ritual of Measuring Height

Preface

Growing up is marked by small moments. For many families, measuring a child's height is one of them. Over time, a wall becomes more than a measuring surface. It becomes a record of shared memories.

But what happens when the family moves, and the wall must be left behind?

This workbook explores how design can help families preserve and continue this familiar ritual, wherever life takes them.

It follows our process through the four phases of the Double Diamond: Discover, Define, Develop and Deliver.

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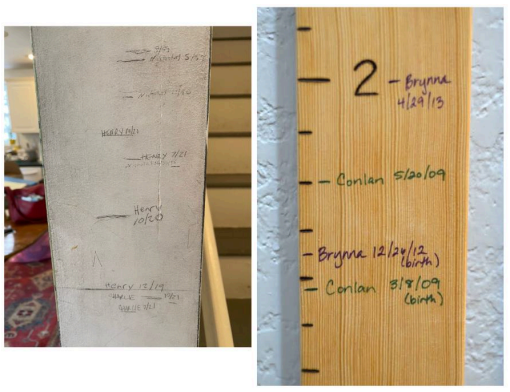
OUR INTERPRETATION

Design Brief

The brief asked us to design for a ritual. We read a ritual as something people do again and again, that leaves a record of a period of life. We compared three everyday practices that fit this reading.



Chosen

A. Height marking

WHO RECORDS

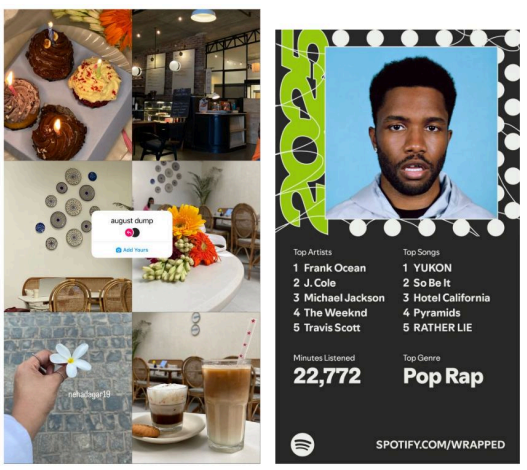
A parent

MEDIUM

The body, on a wall in pencil

CAN IT BE UNDONE?

Effectively permanent



Not chosen

B. Recaps and photo dumps

WHO RECORDS

An algorithm, or the user

MEDIUM

A year of data, a month of photos

CAN IT BE UNDONE?

Deletable any time

Not chosen

C. Film photography

WHO RECORDS

The person, limited by the camera

MEDIUM

Film, prints, album

CAN IT BE UNDONE?

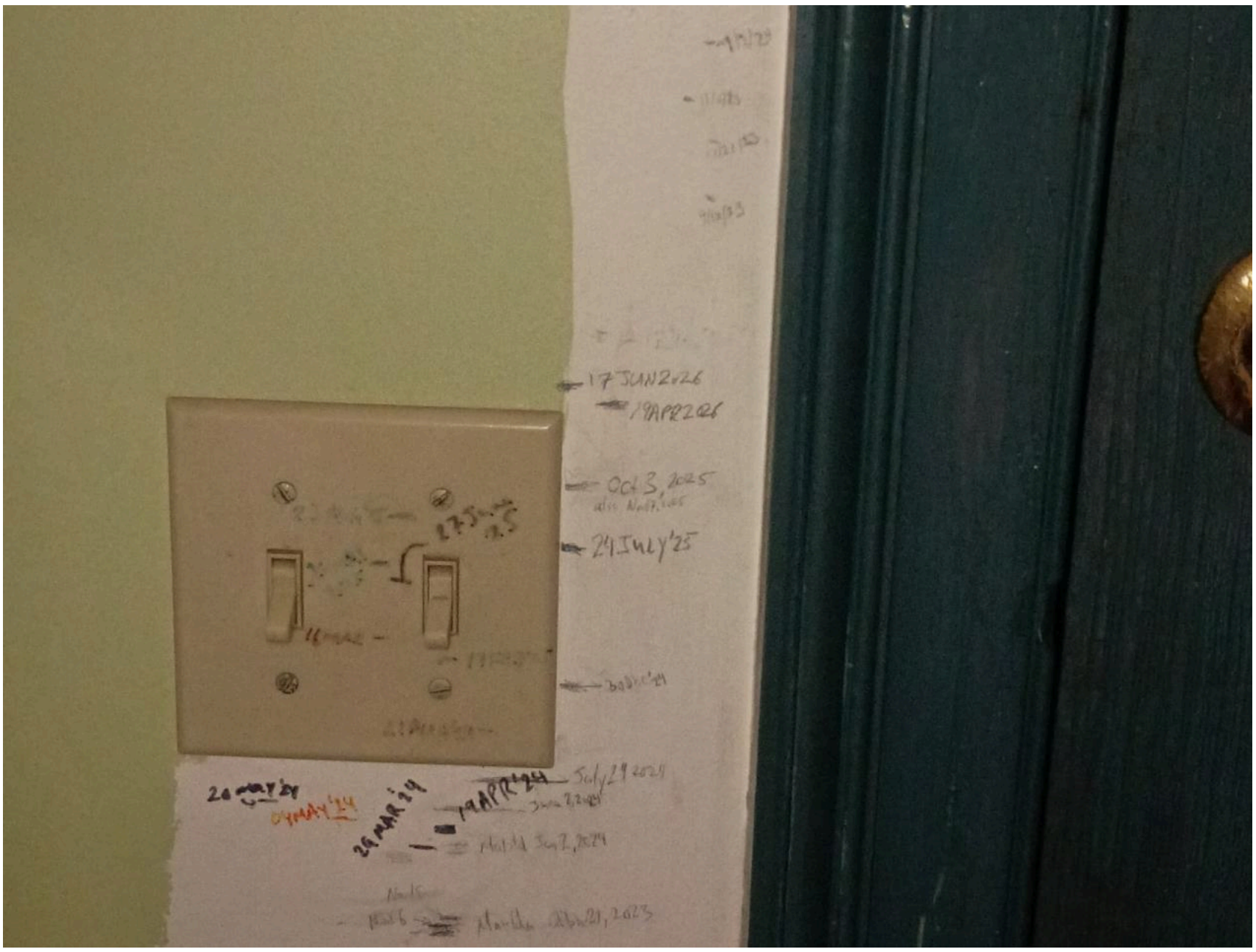
A frame cannot be retaken

WHY WE CHOSE HEIGHT MARKING

Not on a screen

Can't move with the family

Nobody decides when it ends



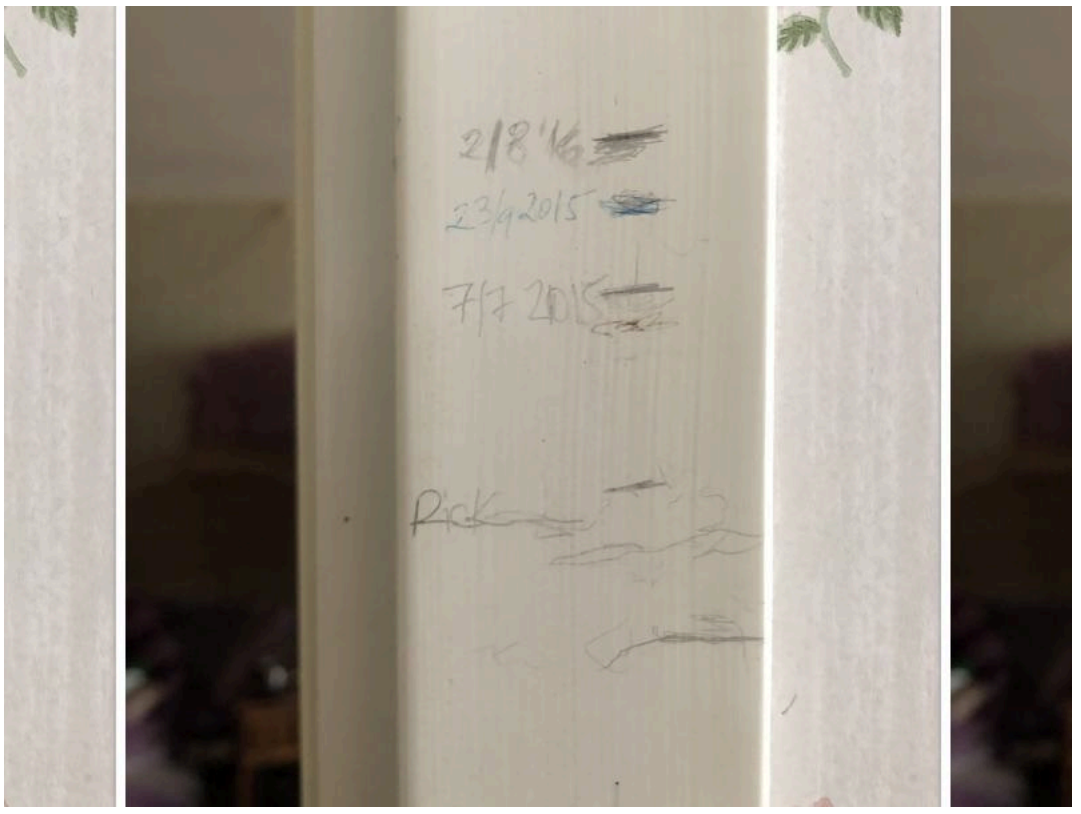
DISCOVER

We talked to families and to adults who grew up with marks on the wall, to learn how the ritual works and how it ends.

Interviews **Drawing task** **Vignettes**

WHO WE TALKED TO

Interviews



We talked to adults who were measured as children and have since moved out, and to parents who still measure their children. Everyone brought a photo of their wall, so the conversation started from a real mark. Interviews in a participant's own language were summarised in English for the team.

SOME THINGS WE ASKED

THE ENDING
"Did you know it was the last time?"

THE MARK
"Who did the measuring?"

PROJECTIVE
"If you could keep only one thing, what would it be?"

SIBLINGS
"Did you compare who was taller?"

THE RECORD NOW
"Is the wall still there?"

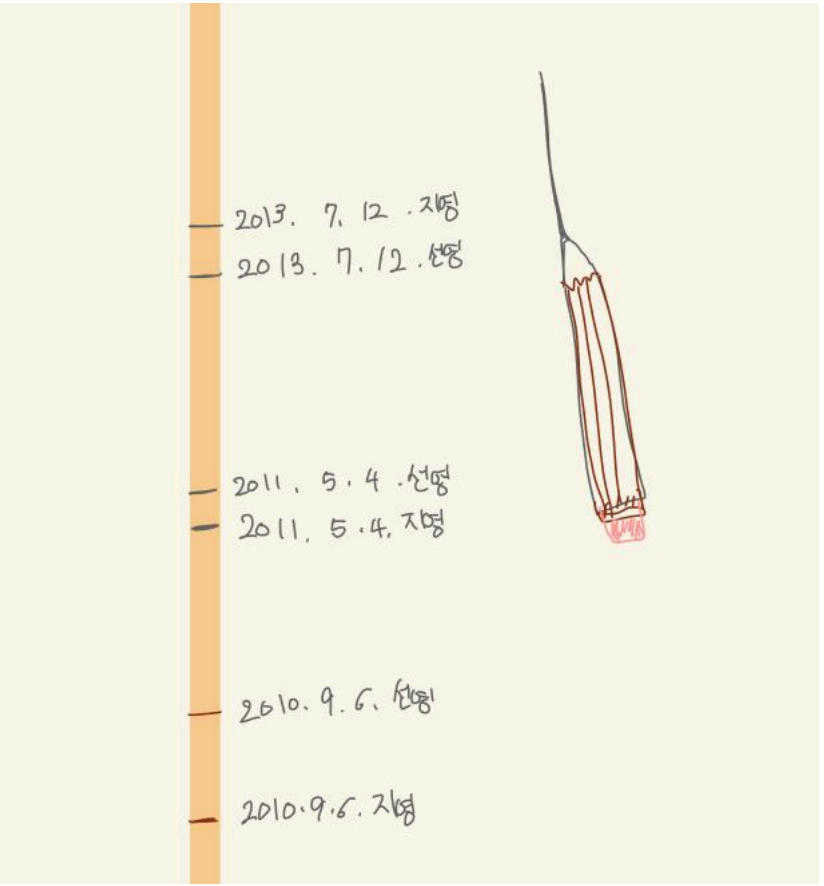
Two of us ran each session: one asked, one took notes.

METHODS

Drawing Task and Vignettes

DRAWING TASK

"Draw that wall the way you remember it." We stayed silent while they drew, then asked about what was on the paper. People remembered details they had not mentioned before: where the marks were, who stood where, what was written.



VIGNETTES

VIGNETTE CARD

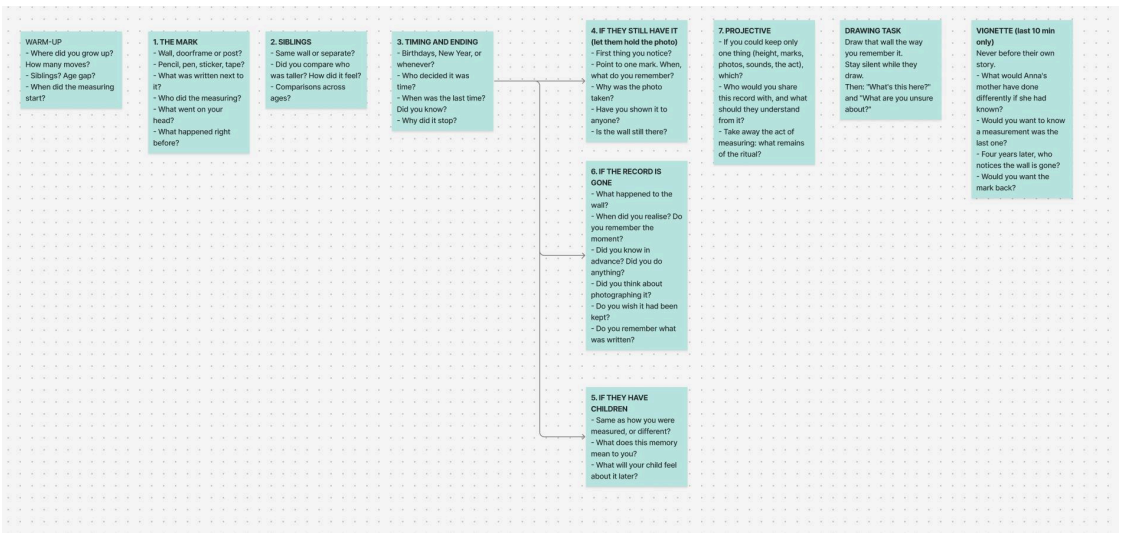
Anna is 14. Every year since she was three, her mother has marked her height on the kitchen doorframe. This year's line is two millimetres above the last. Nobody knows it, but it is the last one. Four years later, the kitchen is repainted.

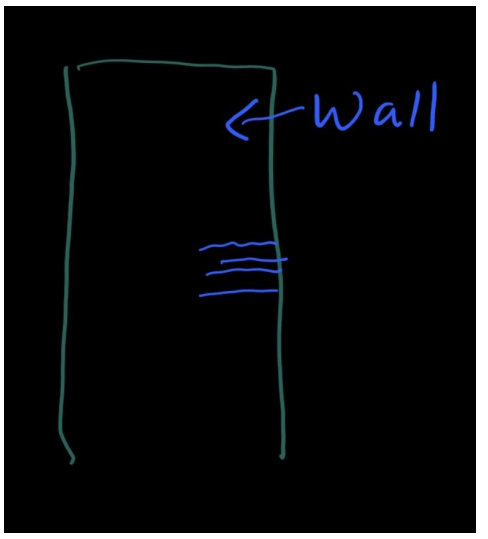
1. **What would she have done differently if she had known?**
2. **Would you want to know a measurement was the last one?**
3. **Four years later, who notices the wall is gone?**
4. **Would you want the mark back?**

We read the vignette in the last ten minutes, after participants had told their own story. Talking about Anna's last mark made it easier to talk about endings and loss, without asking about their own directly.

MAIN FINDINGS

What We Found





01

Nobody plans it

It starts when clothes stop fitting, or someone says "you've grown".

02

Name and date, never age

One pencil line with as little information as possible.

03

Measuring is the fun part

Families enjoy doing it together. Children rarely look at the old lines.

04

Siblings compare

People remember the day one sibling caught up with the other.

05

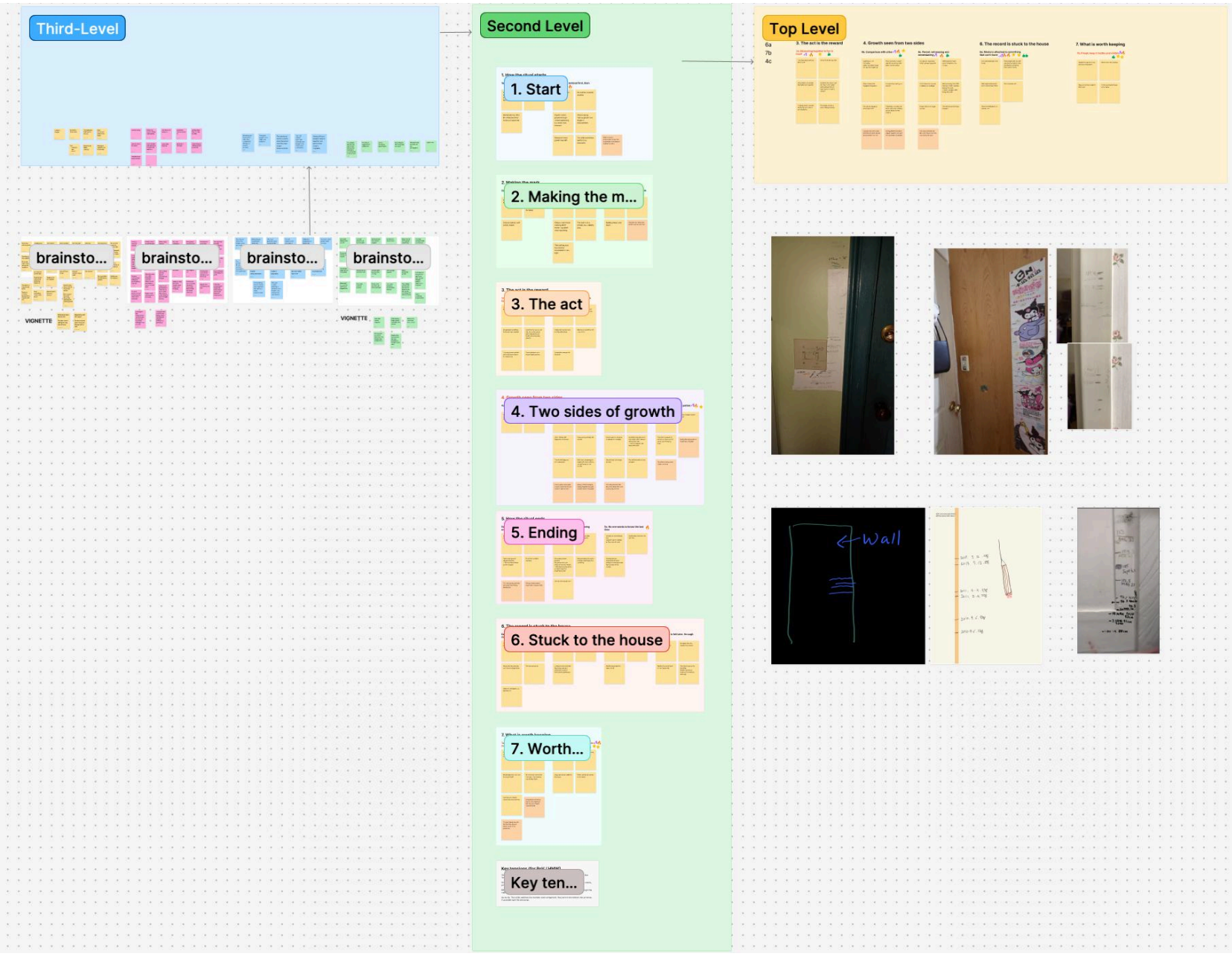
Repainting feels like losing everything

The record is stuck to a wall that can't move with the family.

06

Nobody knows the last time

The ritual fades out. Most would rather not know which mark was the last.



DEFINE

We sorted what we heard, found where it contradicts itself, and turned it into one problem to design for.

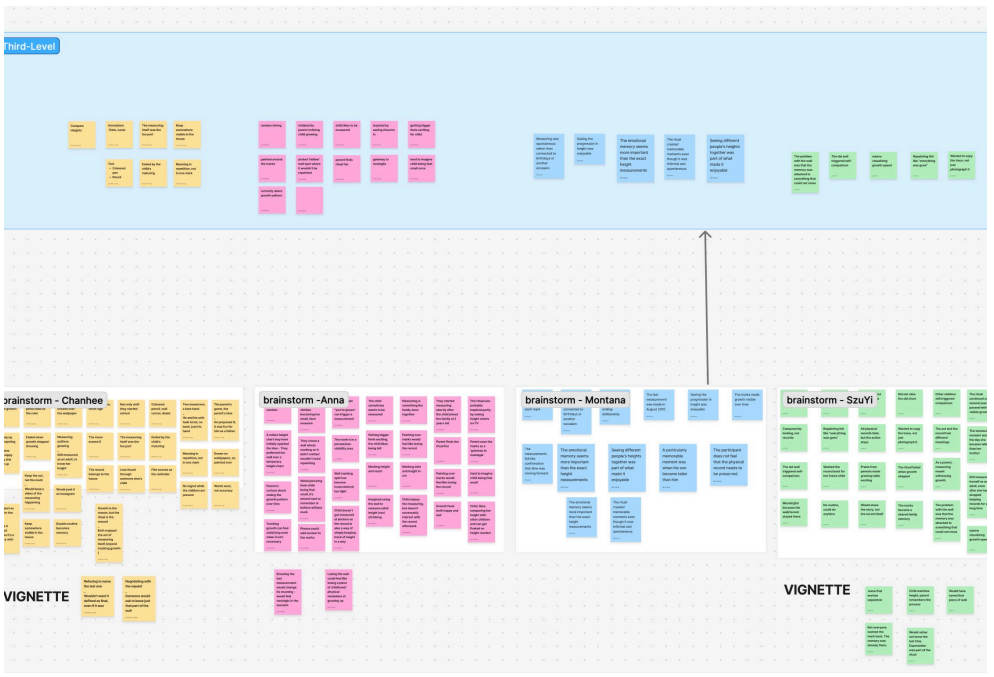
Affinity diagram **Key tensions** **Persona** **POV** **HMW**

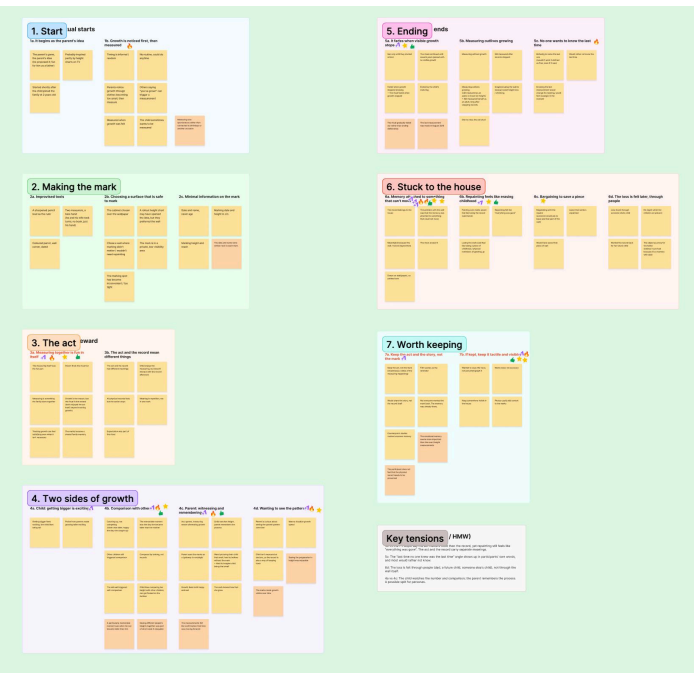
DATA ANALYSIS

Affinity Diagram

Each of us wrote our interview notes on stickies in our own colour, one observation per note. We clustered them bottom-up in FigJam and named each cluster in the participants' words.

1 **Brainstorm** — 2 **Merge** — 3 **Cluster** — 4 **Name the groups**





THE SEVEN GROUPS, WITH ONE NOTE FROM EACH

"Measured when growth was felt"

1 · HOW IT STARTS

"Date and name, never age"

2 · MAKING THE MARK

"The measuring itself was the fun part"

3 · THE ACT IS THE REWARD

"Child watches height, parent remembers the process"

4 · TWO SIDES OF GROWTH

"Would rather not know the last time"

5 · HOW IT ENDS

"The record belongs to the house"

6 · STUCK TO THE HOUSE

"Keep the act, not the mark"

7 · WORTH KEEPING

WHAT THE DATA TOLD US

Key Tensions

The clusters did not point to one problem. They showed four contradictions. We kept them as tensions so we would not jump to a solution too early.

GROUPS 3 VS 6, 7

The act vs the record

People say measuring matters more than the marks. Yet repainting still feels like everything is gone.

GROUP 5

The last mark

Nobody knew which measurement was the last. Most would rather not know.

GROUP 6

Loss is felt through people

People miss a father, or think of a future child. Not the wall itself.

GROUP 4

Child vs parent

The child watches the number. The parent remembers the moment.

FEEDBACK FROM SUPERVISION

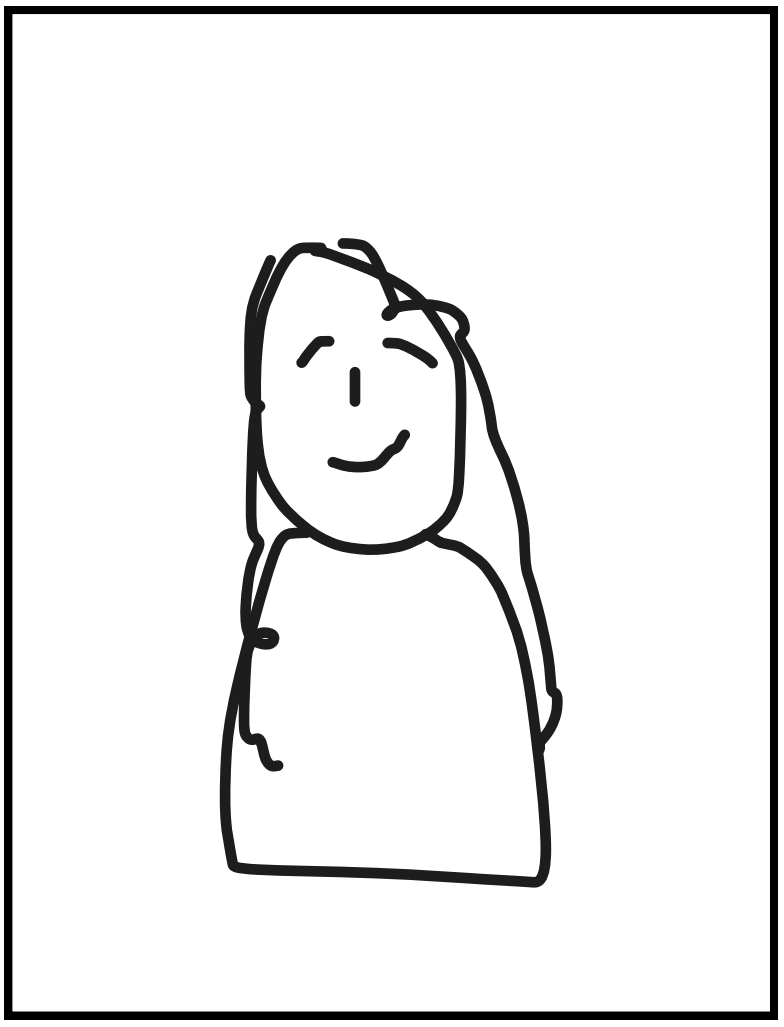
"If you take the measuring away, what is left?"

Our answer became the core of the project: the body itself is the record, and nobody can control their own growth.

WHO WE DESIGN FOR

Persona

Our first interviews were with adults looking back. Their memories kept pointing at families who are still measuring. So we focused on a parent: she starts the ritual, and she is the one who has to leave the wall behind.



Sofia, 36

Mother of two, aged 7 and 4. The family has to move out of their flat next summer.

"It just happens. Someone says they look bigger, or they catch up with a cousin, and suddenly we're at the wall."

WHEN

When clothes stop fitting, or a relative says a child has grown.

HOW

One pencil line on the hallway wall, a name and a date.

WHAT SHE LOVES

Five minutes together, and seeing how much they've grown.

PROBLEM

When they move, the marks stay on the wall.

GOAL

A record she can see every day, not a photo in an album.

PROBLEM DEFINITION

POV and HMW

POINT OF VIEW

A parent who marks her two young children's height on the wall of their home needs the record to survive a move she already knows is coming, because the marks are made on a wall she will have to leave behind while the ritual is meant to last the whole of childhood.

How might we...

...keep a growth record without a fixed place to keep it in?

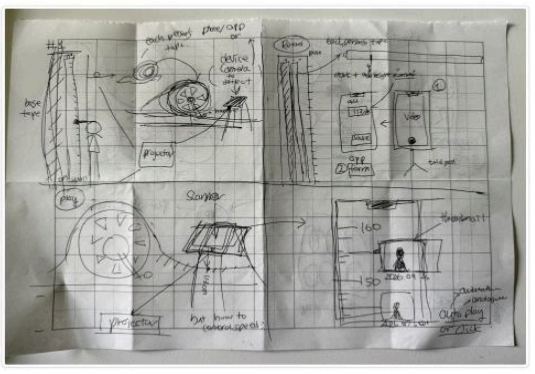
...make moving a step in the ritual rather than its end?

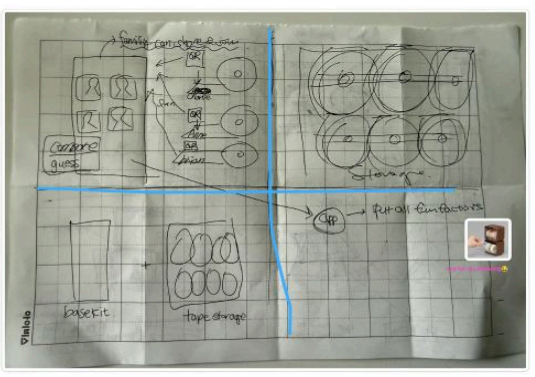
...make comparing siblings' growth a fun part of the ritual?

Direction 1 · Portable ritual

Direction 2 · Sibling comparison

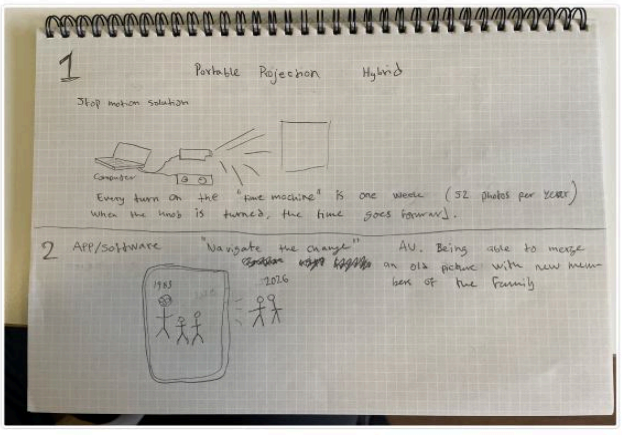
Chanhee



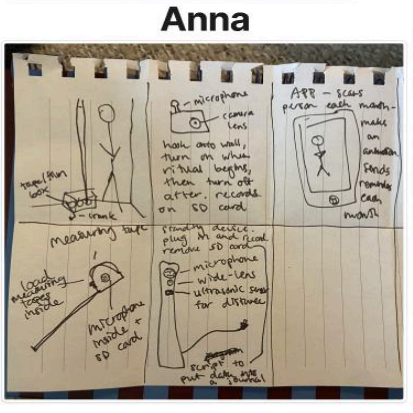


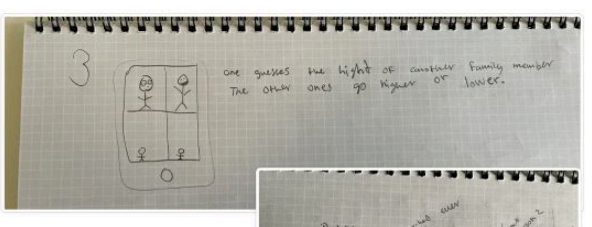
Szu Yi

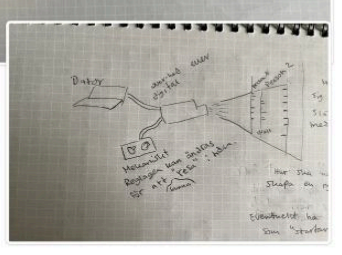
Montana



Anna







DEVELOP

We generated ideas, sketched them, built a first version and tested it with another group.

Brain dump · Brainwriting · Sketching · Worst possible idea · Lo-fi prototype · Peer evaluation

GENERATING IDEAS

Ideation

We ran each method once for each HMW direction. Every round started with writing alone, so the first idea said out loud would not steer everyone else.

1

Brain dump

Every idea, no judging.

2

Brainwriting

Building on each other's ideas in writing.

3

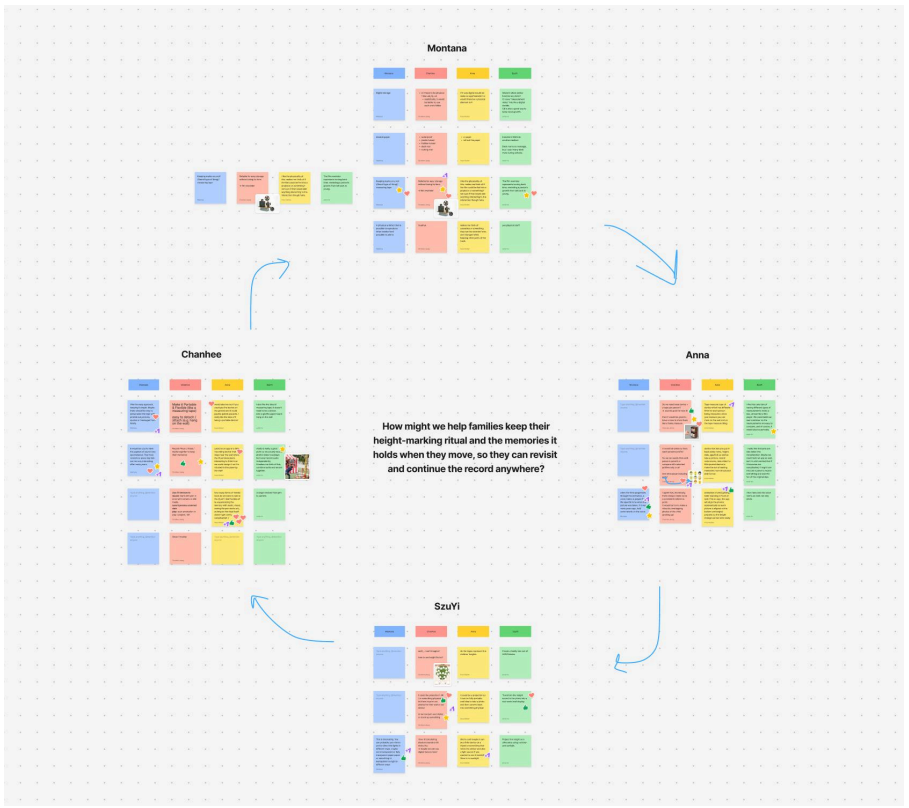
Sketching

The best ideas drawn at a real wall.

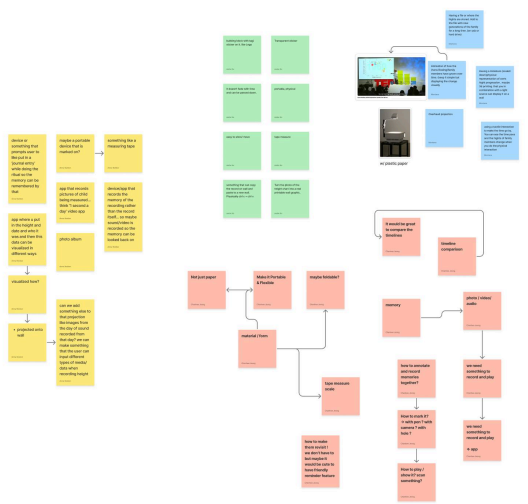
4

Worst idea

Terrible on purpose, then flipped.



Brain write



Brain dump

IDEAS WE KEPT COMING BACK TO

SZU-YI

Copy the record on the old wall and paste it on the new one.

ANNA

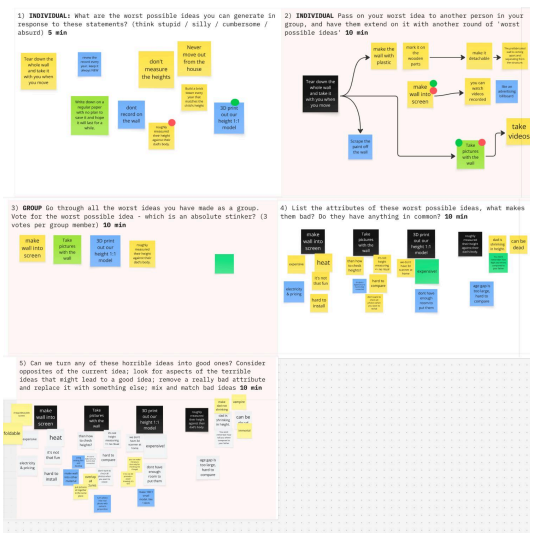
A tape measure with a film for each person.

MONTANA

Silhouettes that move the way the person moves.

CHANHEE

Record the moment with a camera, play it back on the wall.

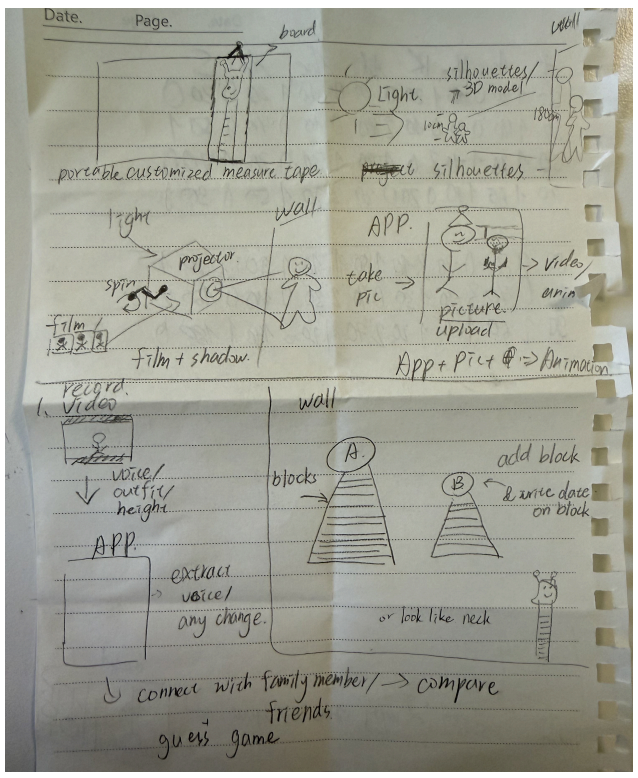


Worst Possible Idea

IDEATION

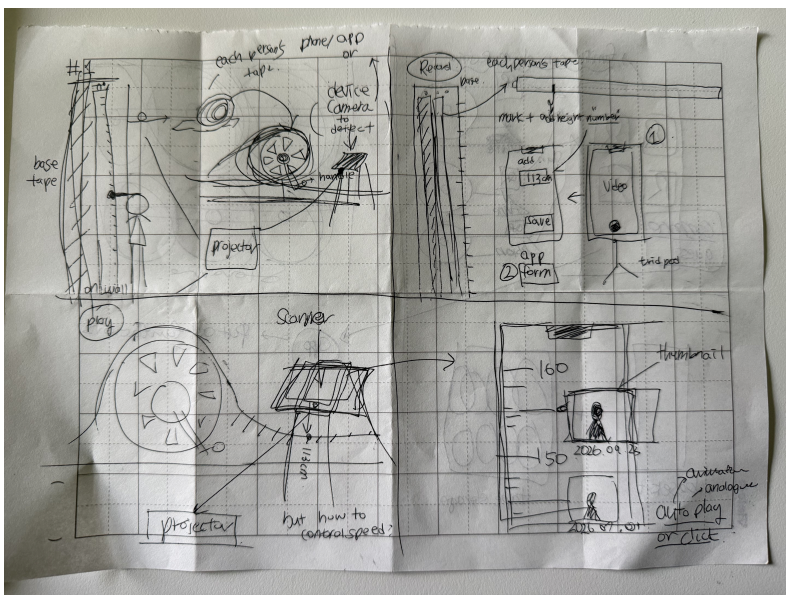
Sketches

After brainwriting, each of us sketched one idea at a real wall. The same themes kept coming back: record the moment, then play it back with light.



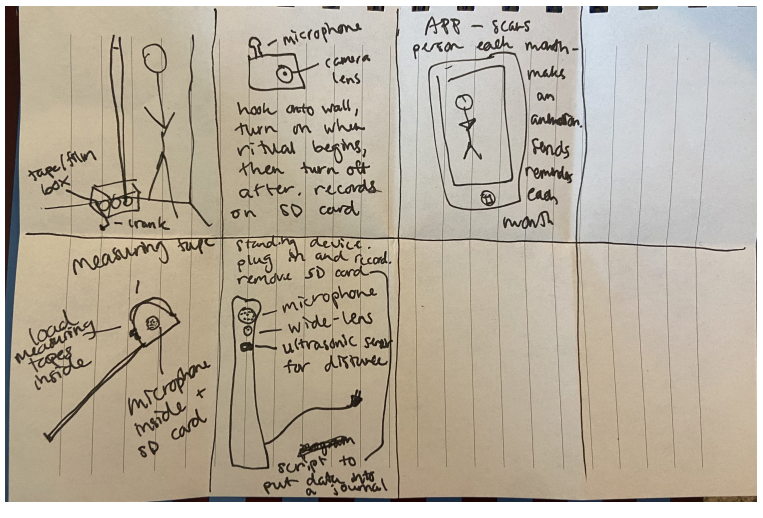
Szu-Yi

Record the moment on video and play it back on the wall with light.



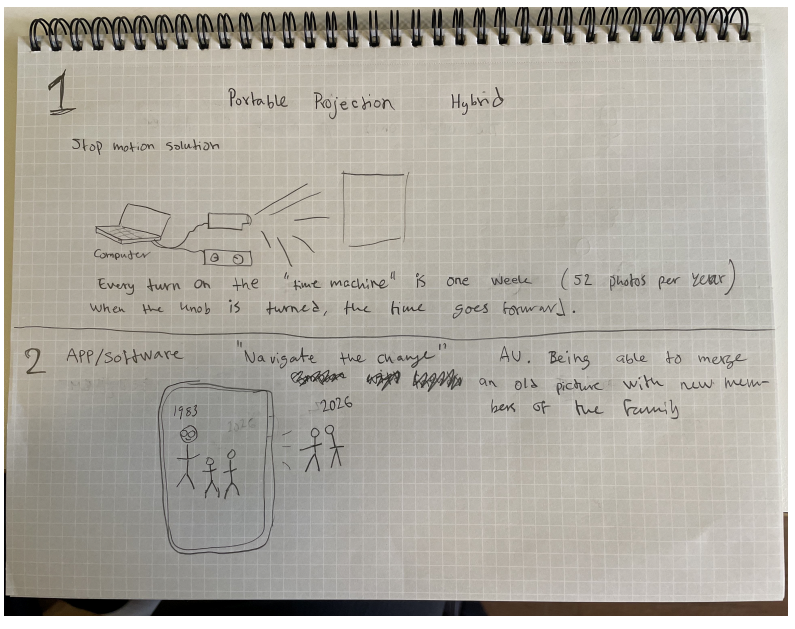
Chanhee

A tape for the marks, plus video and audio, played back with a projector.



Anna

A tape that holds the marks, with a video and voice of that day.



Montana

Silhouettes made with light and shadow. But how does it work without a computer?

CONVERGING

From Ideas to Features

We traced every sketch back to the ideas it came from and wrote each one up as a feature. Then we kept only what answers the problem statement, and drew our scenario as a storyboard to check that the features work together in one story.

#	Feature	Definition	Source ideas
1	Portable physical record ⭐	Marks are made on a tape-like medium (measuring tape) instead of the wall. It can be removed, stored, and taken along when moving	measuring tapes, portable / flexible / easy to store & move, film / tape / storage
2	Video capture of change ⭐	Records a short video at the moment of measurement, capturing changes beyond height such as appearance, voice, fashion, and hair	photo / video / audio / voice, not just height → Change, device → record
3	Growth playback & visualization ⭐	Plays back past videos in chronological order through projection	projection, visualize the growth (timeline, animation), device → play
4	Comparison	Shows the height gap between siblings or family members, or between past self and present self	Height gap, comparison more exciting
5	Guessing game	A playful element where the child predicts their height before measuring, then the result is revealed	Guessing game
6	Companion app 👀	An app for storing and viewing digital records	app, physical + digital + analogue

Core Feature

Scenario Storyboard

WHAT WE KEPT

Video of the moment

A short clip when measuring. It keeps more than height: voice, hair, clothes.

Playback on the wall

Past records shown in order with a projector.

Comparison

Your past self against today, or against a sibling.

Physical controller

So the family looks at the wall, not at a phone.

WHAT WE LEFT OUT

Portable measuring tape

Marks on a removable tape. It faded out as the design took shape and added little next to the other features.

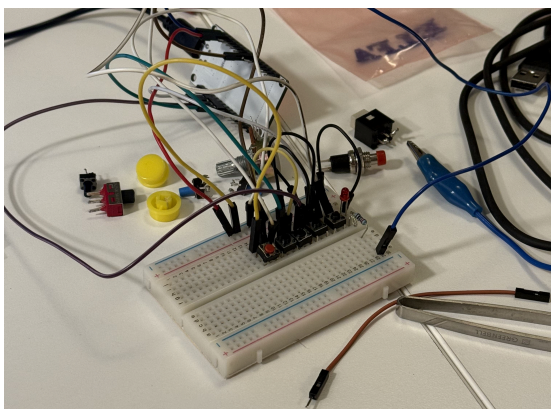
Guessing game

Guess your height, then reveal it. Fun, but lower priority than the core features.

FIRST VERSION

Lo-fi Prototype

We built a quick version to see how a family would stand, film and look back at the wall. [Add: what it was made of.]

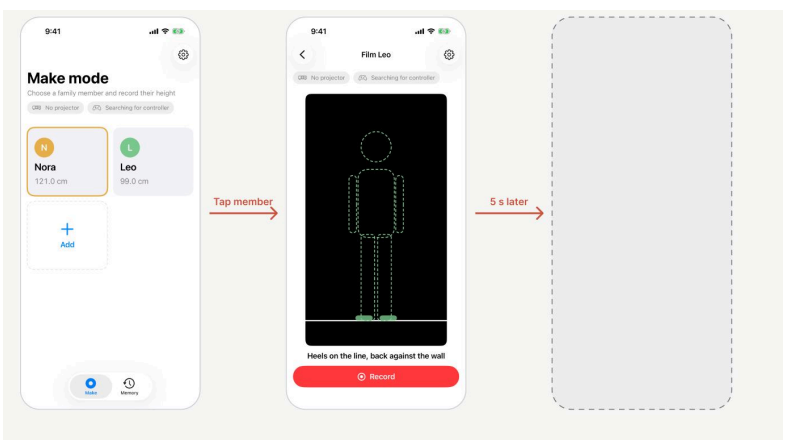


APP WIREFRAMES: TWO MODES

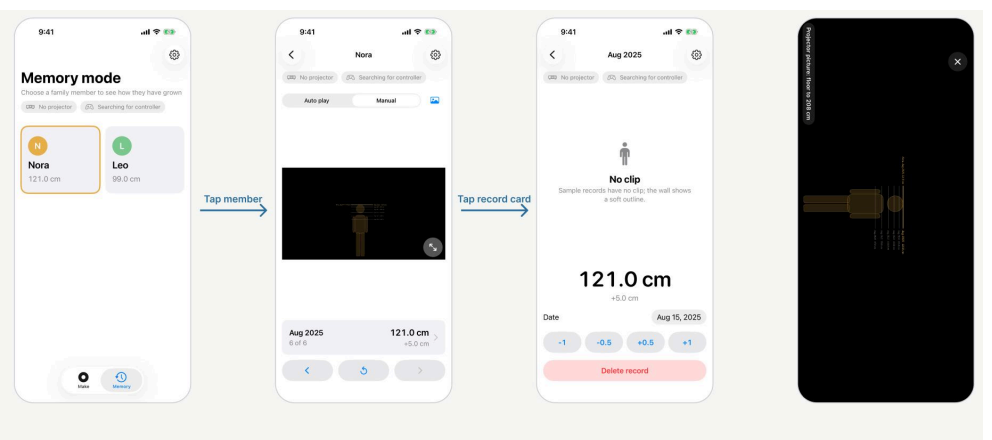
Make | Memory

Choose member → Film → Check height → Save

Choose member → See all records → Step through, or auto play



Make Flow

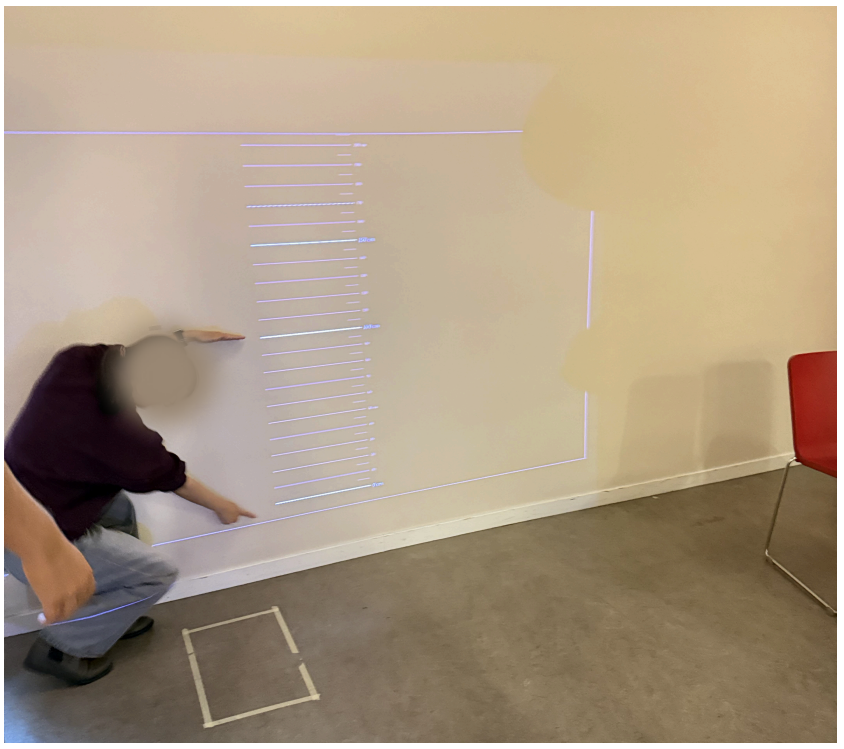


Memory flow

TESTING WITH ANOTHER GROUP · 9 OCT

Peer Evaluation

Another group tested the working prototype with cooperative evaluation: they used it themselves and explained what they were doing, and we could ask questions along the way. Then five of Nielsen's heuristics and a group interview.



WHAT WE NOTED

TEST A · PROJECTOR

Didn't understand why the projector needs setting up at all.

TEST A · WALL

"Floor line" and "size" were unclear. Easier the second time.

TEST A · CAMERA

Thought the camera had to be much closer to the wall.

TEST B · MEASURING

The person squatted to fit inside the old heights.

TEST B · HEIGHT

Could only change the height one centimetre at a time.

TEST B · LOOK BACK

Loved it.

"Playback is really cute."

"Too many pages, too many functions."

"Controller not needed?"

PEER EVALUATION · 9 OCT

Trying It Out

Tasks and when they count as done

Task	Screens	Done when		
A1 Place the projector	S1	Picture touches the floor and reaches at least 170 cm	• didnt understand the task, need to allow permissions • didnt understand the concept of setting the projector	
A2 Fit the picture to the wall	S2	Projected 150 cm is within 1.5 cm of the real 150 cm	• understand the visual concept now of how to set this up • think its cool and interactive • didn't understand 'floor line' and 'size'. need to do size as well, not just floor line • after doing it once its much easier • no back button	
A3 Show the camera the wall	S3	Floor and 100 cm lines placed; the app is ready to record	• thinking the camera has to be a lot closer to the wall •	
B1 Measure your partner	M1 → M2 → M3	Record saved	• person getting measured though they had to squat, the default projection seems like you should fit within the existing heights	
B2 Check the height	M3 or R3	Saved height is within 0.5 cm of the tape measure	• can only add one cm add a time	
B3 Look back	R1 → R2	Finds the partner's record and Nora's oldest and newest records	• loved	

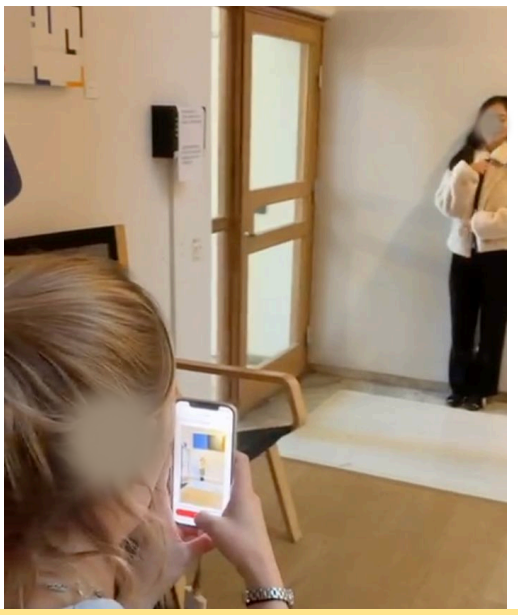
Task-based Usability Testing

Heuristic	notes
Visibility of system statusCould you tell what the app was doing during setup, the countdown, filming and cutting out?	during set up: always disoriented, new interface that she needed to understand playing back the video then everything was understood, add an intro
Match with the real worldDo the setup steps and the marks on the wall make sense in a real home?	yes, maybe we can just skip the setup and focus on taking the photos and then looking back
User control and freedomCould you undo a wrong setting or a wrong record easily?	no, there was **no back button** to change the settings, confusing for how to alter the height after the photo is taken
Recognition rather than recallCould you use the app and the controller without being told what each control does?	no, **controller just felt unnessary** and made the interaction more complicated, would be better without controller
Recover from errorsWhen something went wrong, was it clear what to do next?	hard to identify what went wrong, could get rid of the set up entirely, simplify to the core feature

Nielsen's heuristics

What we tested

We had another group interact with our prototype with no prior knowledge of our concept or design. One at a time, the group members used the prototype and said out loud what they were doing while we observed and asked questions. We then asked them questions we prepared about the usability, intuitiveness, and their experience.



Easier the second time.

Test A, fitting the picture to the wall

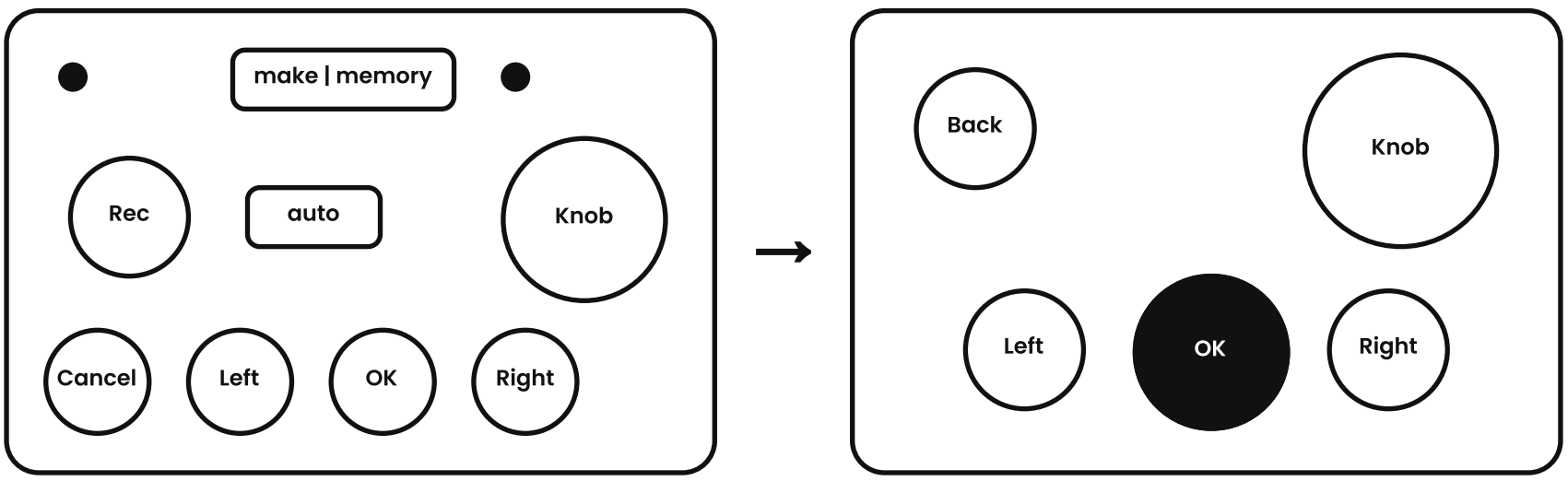
What we learned

Setting up the projector was the hardest part. Looking back at the records was the part testers enjoyed most.

AFTER THE PEER EVALUATION

What We Changed

Testers suggested dropping the controller. We kept it, because looking at the wall instead of the phone is the point of the design. But we removed everything on it that needed explaining.



BEFORE · 9 OCT

AFTER · FINAL

"Controller not needed? Excessive."

So: five controls only. No switches or lights. OK also starts and stops filming.

"No back button."

So: a Back button on the controller.

"Only one centimetre at a time."

So: Left and Right change the height by 1 cm, the knob by 0.1 cm.

"Too many pages, too many functions."

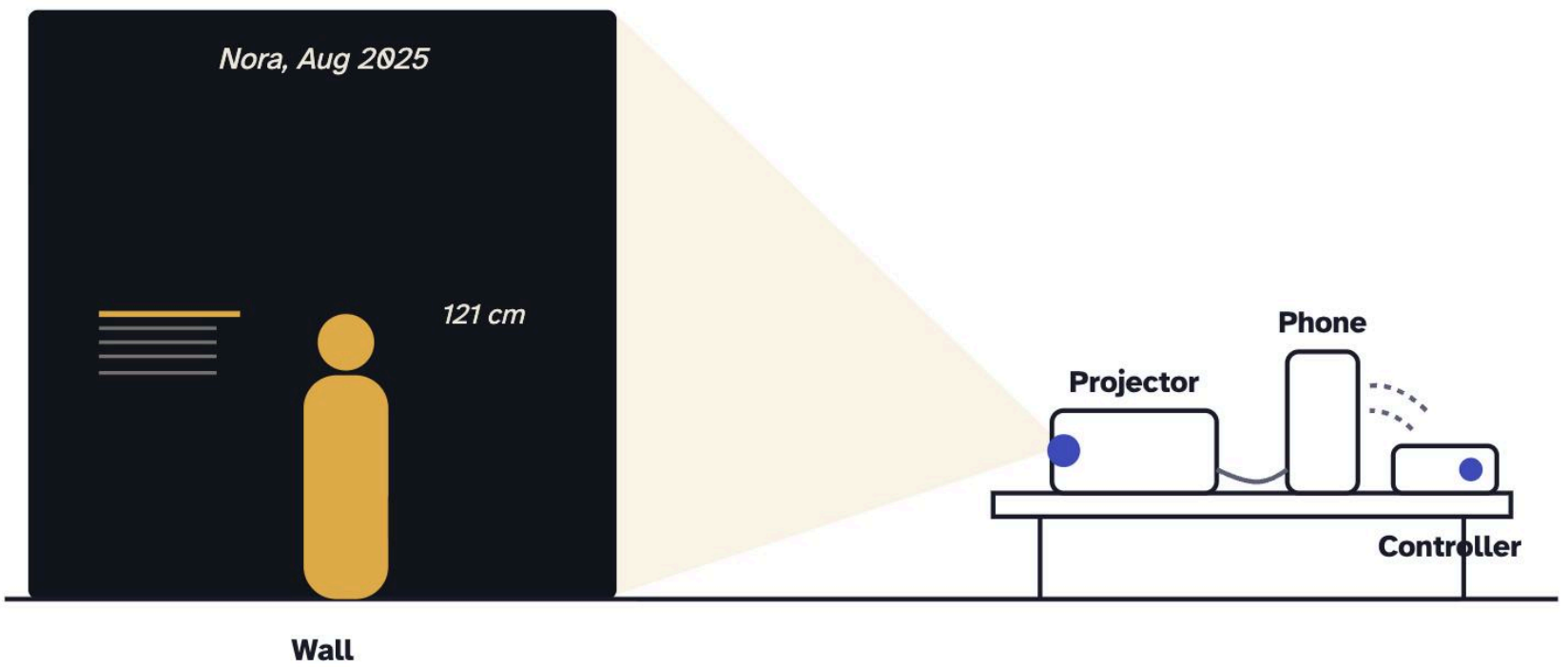
So: we rebuilt the app. Fewer steps, a shorter setup and fewer buttons on screen. Controller icons show which button to press, and a short onboarding explains the rest. The phone and the wall now show different screens, so the family can watch the wall and use only the controller.

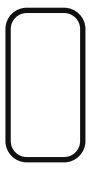
DELIVER

We built the working prototype, an app, a projector setup and a controller, and tested it with users.

Hi-fi prototype **Projector setup** **Controller** **User test**

How It Works



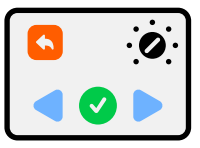


Phone

Films, stores every record, sends the video to the projector.

Projector

Shows the records life-size on the wall.



Controller

Four buttons and a knob, by Bluetooth.

ONE MEASUREMENT

1

Choose who is being measured.

2

They stand on the foot marks. A short video is filmed.

3

The app suggests a height. A parent checks it with a tape.

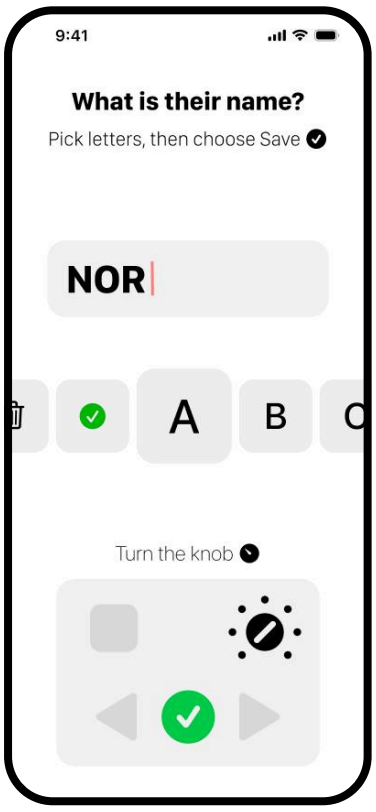
4

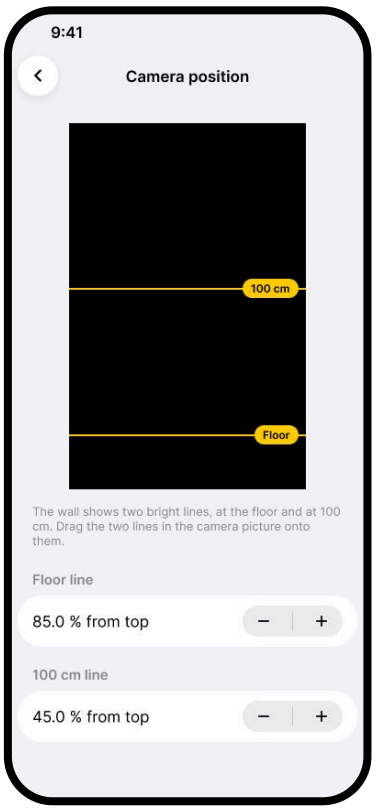
Every record appears on the wall. Turn the knob to move in time.

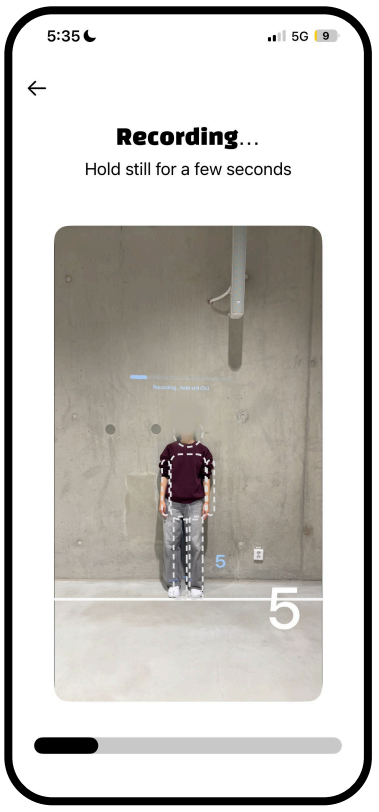
HI-FI PROTOTYPE · APP

Phone Screen

The phone is for the person measuring, and everything on it works by touch. The whole app can also be used on the phone alone, without the controller.







1 · Onboarding

Learn the app and the controller at the same time.

2 · Setup

As simple as possible. Size and distance adjust to fit any home.

3 · Make

Scroll left and right to choose who, then film and save.

4 · Memory

Scroll left and right through every record, the same move as on the controller.

Touch first

Every step is a tap on a large target. Nothing small to aim at.

One step at a time

One question and one obvious next step, because the setup is already hard.

Works without the controller

Every step can be done on the phone alone, so the controller is a choice.

Touch on the phone. Controller on the wall.

The two screens show the same moment, but each is made for the way it is used.

HI-FI PROTOTYPE · APP

Wall Screen

The wall screen runs from the app on the phone, but it is built around the controller. Nobody touches it: everything on it is designed to make the controller as easy as possible, so it always shows where you are and which button to press.

1

Cursor and highlight

A cursor and a highlight show what is selected right now, so nobody has to look at the phone.

2

Button shapes on screen

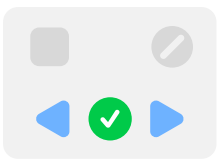
Each button on the wall shows the shape of the controller button that presses it, Back included.

3

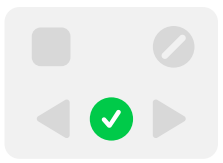
Why we added them

In the peer evaluation, testers struggled with the controller and missed a back button. The guides make it easier to use.

WHICH BUTTONS WORK, AS SHOWN ON THE WALL



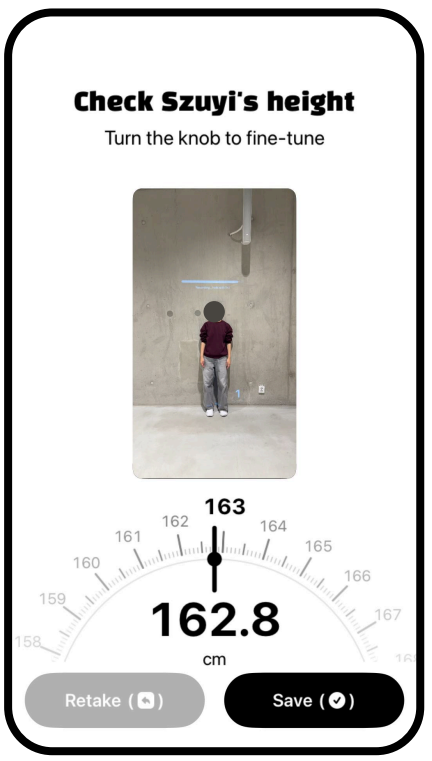
Choosing
Left, Right, OK

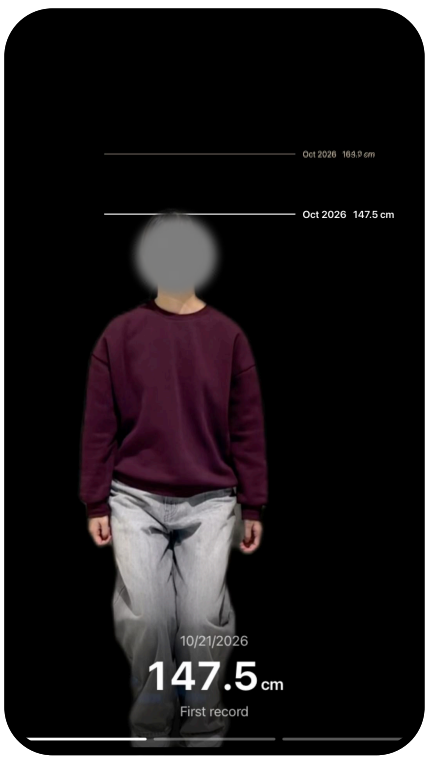


Filming
OK starts and stops

Height
Knob 0.1 cm, Left and Right 1 cm, OK saves

Memory
Turn to move in time



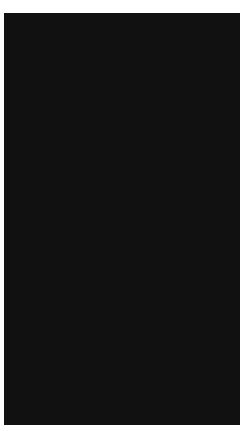


HI-FI PROTOTYPE · INSTALLATION

Projector Setup

We had an ordinary projector, not a short-throw one, and we did not want to mount it on the ceiling. A life-size person needs a picture that starts at the floor and reaches above head height. So we turned the projector on its side.

Upright
A wide picture. It has to stand far back.



On its side
A tall picture, the shape of a person.

THE APP WALKS YOU THROUGH IT

1

Place the projector

The app says how far from the wall. The bottom of the picture touches the floor.

2

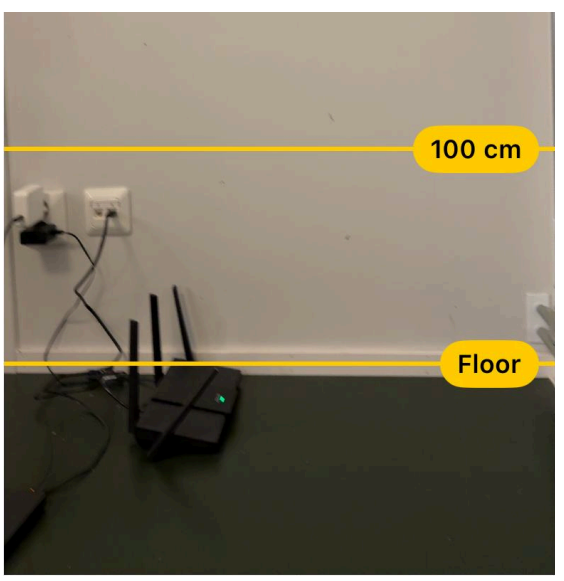
Fit to the wall

Tape at the floor and at one metre. Move and resize the ruler until it matches.

3

Show the camera

Drag two lines onto the projected lines, so the app knows where the floor is.



HI-FI PROTOTYPE · HARDWARE

Controller

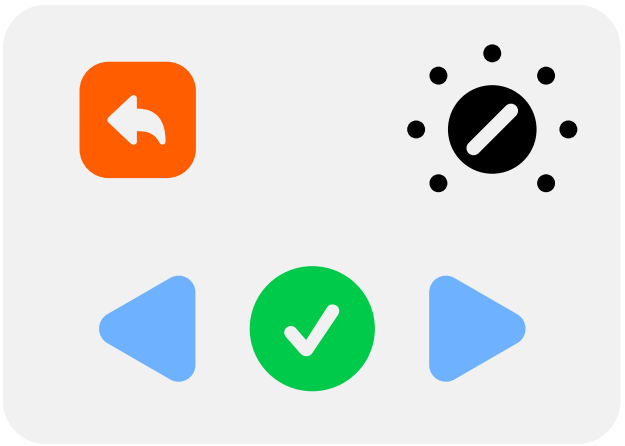
The controller lets the family look at the wall instead of the phone. Inside is an ESP32 that sends button and rotary encoder data to the phone via Bluetooth. The app decides what each press means on each screen.

STEP 1
Sketch

STEP 2
Stripboard

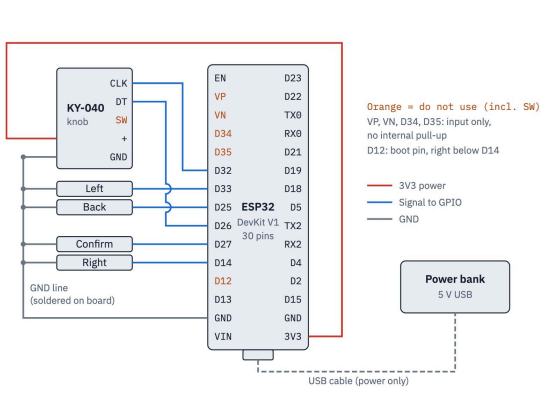
STEP 3
3D print

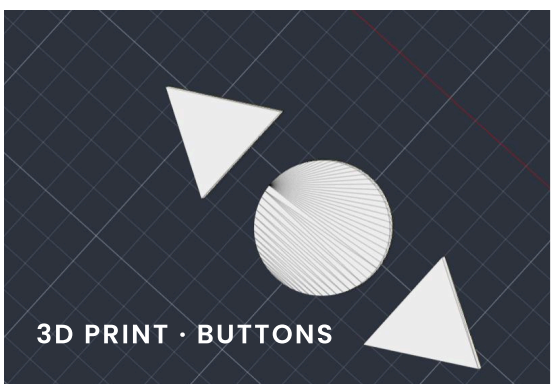
STEP 4
Assemble

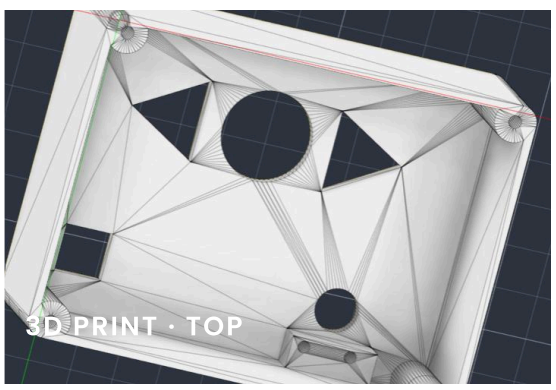


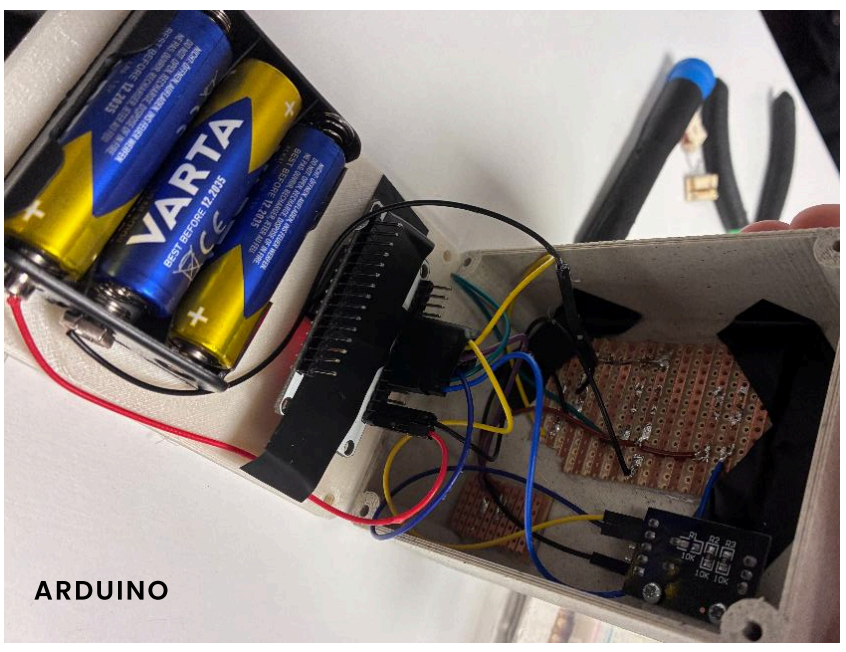
FINAL LAYOUT

Back top left, knob top right, then Left, Confirm and Right. Confirm also starts and stops filming. Powered by a 3 AA battery pack.



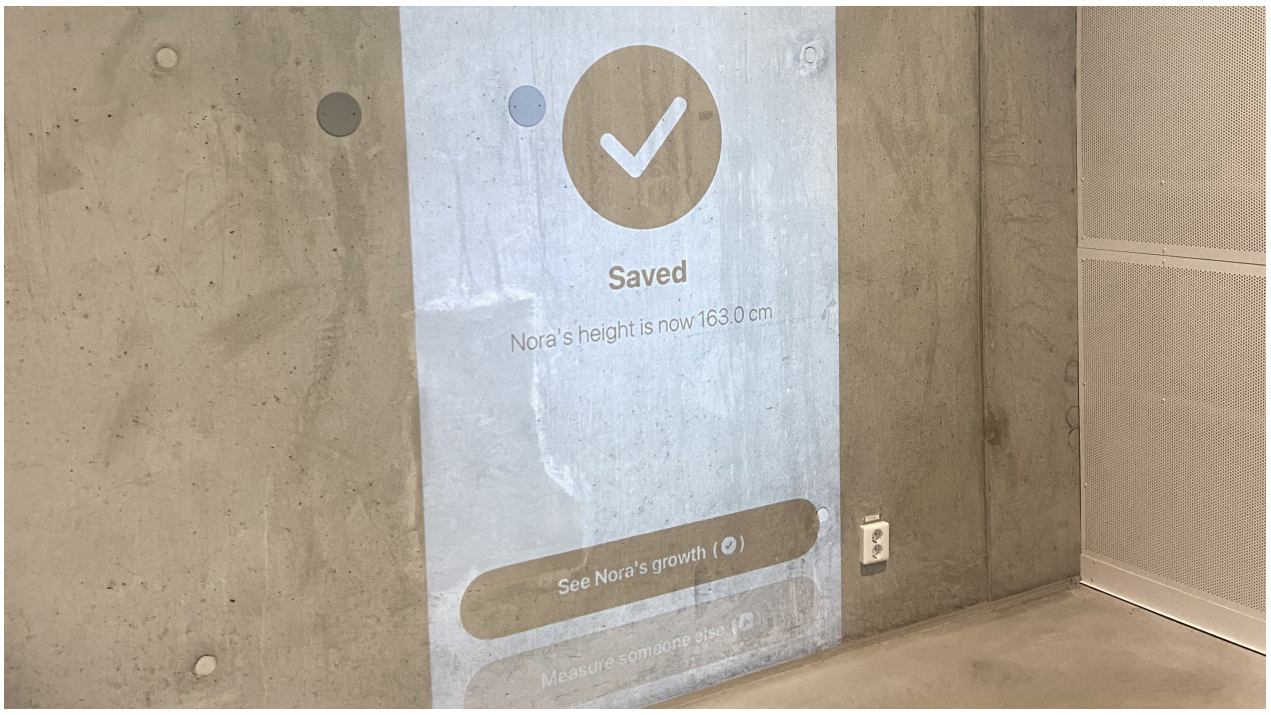






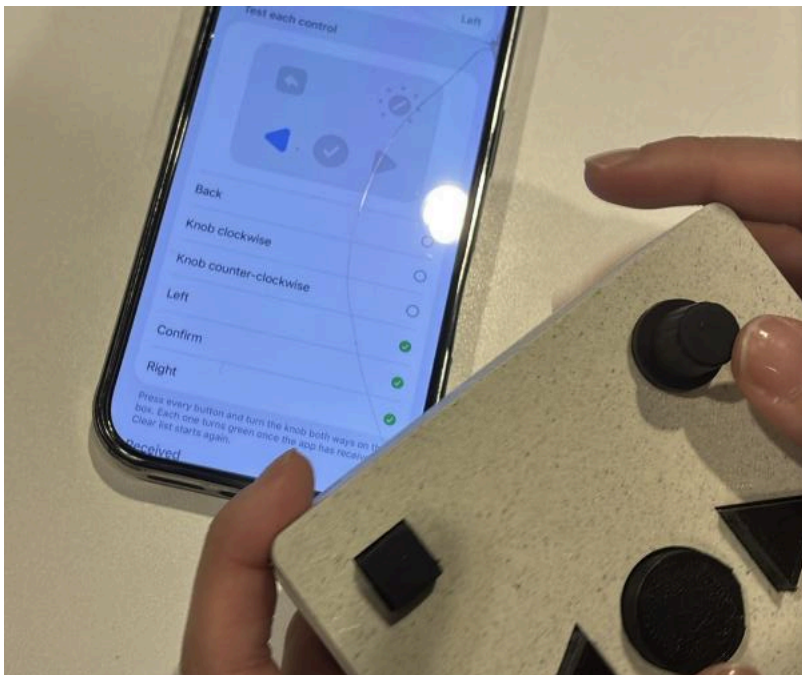
HI-FI PROTOTYPE

Building the Hi-fi



First light

The first time the app's wall image hit a real wall. From here we could check sizes in real centimetres and adjust colours to suit projection.

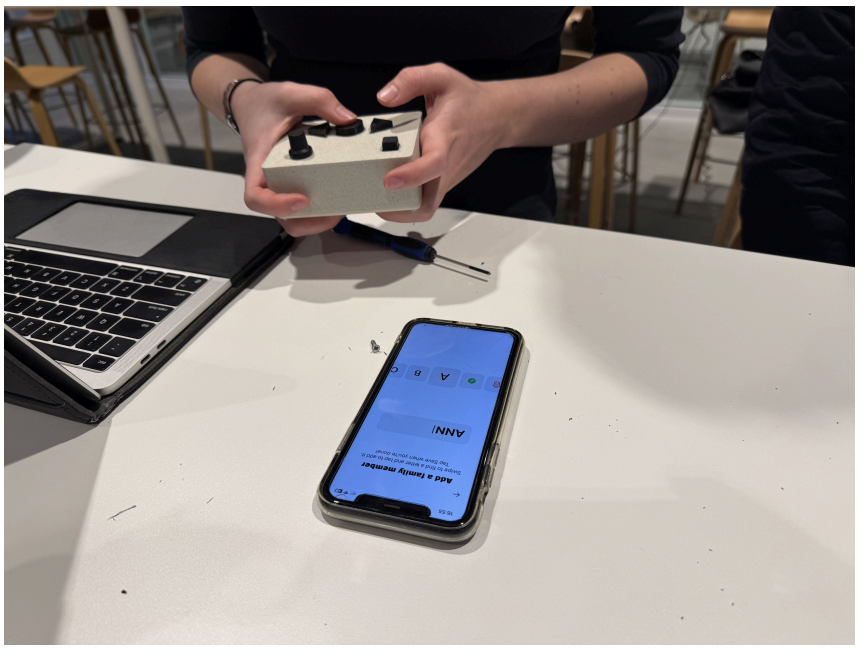


The controller connected to app

We used an ESP32 to connect the controller to our app via Bluetooth and tested whether each button functioned correctly.

Projector projects App’s screen on the wall

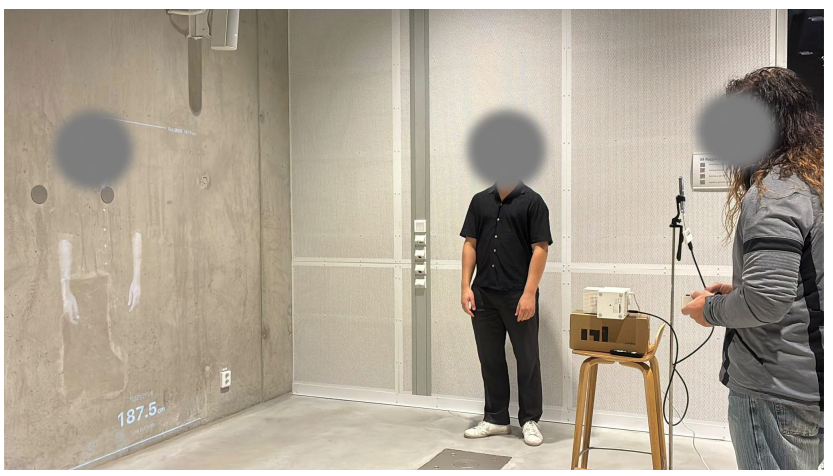
A base, a top plate and four button caps, printed and put together around a battery pack.



TESTING WITH FAMILIES

User Test

We tested the hi-fi prototype at KTH campus on 21 October 2026, with a teenager and a father of adult children. We used the same consent form as in the interviews.



What we asked them to do

Make a record with the controller, then look back at it on the wall.

What worked

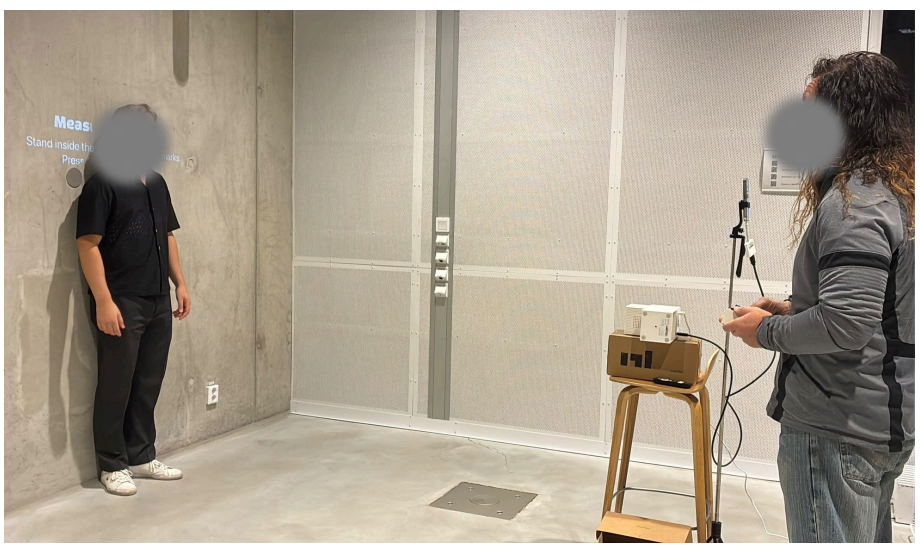
Both enjoyed using it. They found the app easy and intuitive, and the controller easy to adjust.

What they said

They called it straightforward and fun. The father said he would use it from the time children are small.

What we'd change next

Testers suggested other kinds of growth, such as a pregnancy: recording a growing belly and looking back at it later.



"easy to use the controller"

"Straightforward, and fun."

FINAL DESIGN CONCEPT

Memory Meter

Memory Meter keeps the act of measuring and makes the record portable. The body is still the record. The wall is only where it is shown.

WHAT THE PROTOTYPE DOES

Records the moment

A short video every time the family measures, with the height and the date.

Shows it life-size

Past records are projected on any wall at their real height.

Moves with the family

Everything is stored on the phone, so the record survives a move.

FOR OTHER STUDENTS AND DESIGNERS

Takeaways

DISCOVER

Bring the object into the interview

A photo or a drawing of the wall got stories that questions alone did not.

DEFINE

Keep tensions as tensions

Our direction came from a contradiction, not a pain point.

DEFINE

Ask what is left without the obvious part

Taking the measuring away showed us what the ritual really records: the body.

DEVELOP

Find your hardware limits early

Our projector changed the shape of the device more than once.

DELIVER

Testers show you what to cut

They loved the playback and found the rest in the way. Both halves were useful.

MEMORY METER

THE WALL STAYS. THE MARKS COME WITH YOU.

Memory Meter records a short video each time a family measures a child's height, and projects every record back onto any wall, life-size, wherever the family lives next.

Group 6 · Anna Nolden, Chanhee Jeong, Montana Eriksson, Szu-Yi Ho

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KTH Royal Institute of Technology · 2026